Paper	Year	Organism	Variables	ODE	PDE	Other models	N. eq.	N. / type of eq.	Spatial dynamics	Aims, questions	Relevant parameters	Mathematical analysis	Numerical methods / simulations	B.C.	1D	2D	Turing instab.	Spatial patterns	Dynamics	JC	Results
Bonanomi et al., 2005: Negative plant-soil feedback and species coexistence	2005	Plants	N/A	no	no	IBM	N/A	N/A	N/A	NPSF can prevent monoculture dominance, accelerate plant succession, and explain stable species coexistence?	f: NF intensity parameter. d: Dispersal distance m: Plant mortality rate. c: Competitive capability	N/A	Java: stochastic MonteCarlo, discrete spatial grid	periodic	no	yes	N/A	Aggregate multi-species patches	Cyclic population oscillations	no	NPSF : no competitive exclusion but long-term stable coexistence with emergent positive reciprocal interaction
Bonanomi et al., 2006: Phytotoxicity dynamics of decaying plant materials	2006	Plants - leaves and roots	N/A	N/A	N/A	Experimental / Decomposition dynamics model	N/A	N/A	N/A	Investigate the dynamics of phytotoxicity in decaying plant materials across different functional groups over time.	N/A	Statistical Analysis (ANOVA)	Data-driven empirical modeling.	no	N/A	N/A	N/A	N/A	N/A	no	Phytotoxicity general phenomenon, different dynamics in aerobic and anaerobic conditions
Mazzoleni et al., 2010: Modelling the effects of litter decomposition on tree diversity patterns	2010	Plants	Biomass (B_i) Litter (L_i) Nutrients (N) Autotoxicity (A_i)	yes	no	no	4	4 ODE	N/A	litter decomposition drives an intra-specific NF functionally linked to tree diversity gradients across latitudes.	E: environmental condition	Non-linear ODEs tracking resource waste dynamics	Simulation experiments comparing system behavior with and without NF. Validation with field data	no	N/A	N/A	no	Homog. State	Stationary	no	Without NF: Biomass depends strictly on environmental conditions (low diversity). With NF: Species richness increases from boreal to tropical zones. High species diversity when litter decomposition is rapid (high NF prevents dominance) and high species dominance when decomposition is slow or NF is removed.
Cartenì et al., 2012: NPSF explaining ring formation in clonal plants	2012	Plants - Clonal plants	B, T	yes	yes	no	2	1 ODE (T) 1 PDE (B)	diffusion of B (DB)	Role of NF for ring formation	a: plant basal mortality rate relative to growth rate. Beta, gamma	LSA: 2 eq (bare soil, homog coverage) Bifurcation parameter	N/A	no	no	yes	no	ring formation	invasion fronts TW	no	Travelling Fronts leading to homog. coverage. Circular invasion Inside B reduced by NF
Marasco et al., 2013: Water limitation and NPSF explain vegetation patterns along rainfall gradient	2013	Plants - Roots+ shoots	B, W, T	yes	yes	no	3	2 PDE (B,W) 1 ODE (T)	diffusion B diffusion W (DB, D(W-bB) (Klausmeier on flat terrains)	Pattern due to interaction W-T	s: sensitivity of plants to toxicity T. k: decay rate for T. p: precipitation rate	LSA, stab depends on r (k)	I.C: B ≠ 0 for some random nodes.T=0. W ≠ 0.	Neumann	no	yes	no	spots stripes	stationary	no	s=0 stable patterns. As p increases from bare soil to homog coverage through spots, lab, gaps. s>0: low k B escape from T. high k transition from one pattern to another shifted to higher p
Marasco et al., 2014: Vegetation pattern formation due to interactions between water availability and toxicity in PSF	2014	Plants	B, W, T	yes	yes	no	3	2 PDE (B,W) 1 ODE (T)	diffusion B, diffusion W (DB, DW) (Klausmeier on flat terrains)	Pattern due to interaction W-T	s: sensitivity of plants to toxicity T. k: decay rate for T. p: precipitation rate	LSA: 3 eq. For s>0 Bifurcation analysis: Turing bif.	FD FE. I.C: B ≠ 0 for some random nodes. T=0	Neumann	no	yes	yes for some value of p	spots stripes lab	oscillatory TW	no	s=0 stable patterns. s>0 dynamic patterns, for low k B escape from T. Asymmetric distribution of B (Tail)
Mazzoleni et al., 2015: A novel process-based model of microbial growth: self-inhibition in Saccharomyces cerevisiae aerobic fed-batch cultures	2015	Fungi - Yeast	Biomass Nutrient Inhibitor	yes	no	Process-based model (SD)	7	7 ODE	N/A	Investigate self-inhibition mechanisms in S. cerevisiae aerobic fed-batch cultures using a mechanistic, process-based approach.	N/A	Simulations, statistical analysis, model validation with data from literature and experiments	SIMILE + Matlab: ode15s solver	no	N/A	N/A	N/A	N/A	N/A	no	NF processes important to understand microbial growth processes
Mazzoleni et al., 2015: Inhibitory and toxic effecs of ex self-DNA. A mechanism for NPSF?	2015	Plants	Biomass phytotoxicity autotoxicity	N/A	N/A	Litter collection and decomposition experiments	N/A	N/A	N/A	Significance of autotoxicity. Test if ex-self-DNA induces NPSF	N/A	Statistical analysis of experimental growth curves	Non-linear regression and dose-response curve profiling.	no	N/A	N/A	N/A	N/A	N/A	no	ex -self-DNA inhibitory effects explain autotoxicity and contribute to PSNF
Mazzoleni et al., 2015: Inhibitory effects of extracellular self-DNA: a general biological process?	2015	6 taxonomic groups	6 taxonomic groups. Ex- self-DNA	no	no	Comparative experimental model (exposure to self vs. heterologous DNA)	N/A	N/A	N/A	growth inhibition induced by extracellular self-DNA across different taxonomic groups?	N/A	Statistical analysis of experimental growth curves across kingdoms.	Comparative growth kinetics monitoring and biomass reduction scaling.	no	N/A	N/A	N/A	N/A	N/A	no	ex-self-DNA inhibition universal biological law governing NF. The inhibition is strictly species-specific (autotoxic) and acts as an evolutionary mechanism for self-regulation.
Vincenot et al., 2016: Spatial Self-Organization of vegetation subject to climatic stress. Insights from a system Dynamics-Individual-Based Hybrid Model	2016	Plants	B_i Water (W, O)	yes	yes	SD- IBM	3	1 ODE (B_i) 2 PDE (W, O)	diffusion W	Provide insights into vegetation patterns formed under climatic stress.	N/A	N/A	JAVA: 2 simulation experiments exploring climate change scenarios.	no	N/A	N/A	no	N/A	N/A	no	Spatial self-organization emerges under climatic stress, driven by the feedback loops between individual plant behavior and localized water availability.

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Vincenot et al., 2017: PSNF explains vegetation dynamics and patterns at multiple scales	2017	Plants - Individual plants	N/A	no	no	IBM	N/A	N/A	N/A	Effect of PSNF on spatio temporal dynamics of pop and communities of plants of diverse growth form	level of establishment sensitivity to NPSF. K: decay rate of toxicity	N/A	3 sets of simulations: seed dispersal, Vegetative propagation, short-lived herbaceous plants with seed dispersal	no	no	yes	no	Localized patches	N/A	yes	PSNF: driver of JC distributions. A)JC distr. High PSNF: no seedling under crow. B) no other plants: spreads in circular fashion. With PSNF toxicity accumulate in centre. Interactions 2 plants: initially 2 distinct rings then 1. C) PSNF disfavour individuals of a species to germinate and grow: fast rotation.
Marasco et al., 2020: Modelling competitive interactions and PSF in vegetation dynamics	2020	Plants	B_1, B_2, T_1, T_2	yes	yes	no	4	2 ODE (T_1, T_2) 2 PDE (B_1, B_2)	diffusion B1, B2	Role of Inter and intra specific NPSF	K: ratio between the interspecific toxicity and autotoxicity	LSA Bifurcation analysis: stability depends on K. Hopf and transcritical bif.	Matlab: ode15s solver and FD (FTCS for B, FE for T).	Neumann	no	yes	no	localized patches	Stationary Oscillatory	no	T=1: coexistence, symm. Oscillations. NO stable config.. T=1.5 coexistence, spiral patterns and waves. K=0 competitive exclusion
Cartenì et al., 2020: A general process-based model for describing the metabolic shift in microbial cell cultures	2020	Bacteria - Escherichia coli Bacillus Subtilis	N/A	yes	no	Process-based model (SD)	7	7 ODE	N/A	1) Overflow metabolism represses respiration. 2) NF on growth rate and accelerated cell death due to a generic internal inhibitory factor	N/A	Simulations, statistical analysis, model validation with data from literature and experiments	Matlab: ode15s solver Minimizing SSE Sensitivity analysis	no	N/A	N/A	N/A	N/A	N/A	no	The model effectively simulates the transition into the decline phase and increased mortality driven by by-product self-inhibition.
Iuorio, Veerman, 2021: The influence of autotoxicity on the dynamics of vegetation spots	2021	Plants	B, W, T	yes	yes	no	3	2 PDE (B,W) 1 ODE (T)	diffusion B diffusion W (DB, D W) (Klausmeier on flat terrains)	Pattern due to interaction W-T	s: sensitivity of plants to toxicity T. k: decay rate for T.	LSA Pulse solutions via GSPT	FDFE, I.C: T=0.	not relevant	yes	yes	no	spots lab gaps homog coverage	TW	no	2D: s=0 stable patterns. As p increases from bare soil to homog coverage through spots, lab, gaps. s>0: low k B escape from T. high k transition from one pattern to another shifted to higher p. 1D: H=0 static periodic spots. H>0 Travelling pulses, asymmetric, stable
Chiusano et al., 2021: Arabidopsis thaliana response to extracellular DNA: self vs non-self exposure	2021	Plants - Arabidopsis thaliana	Biomass Self-DNA, Non-Self DNA	N/A	N/A	Experimental / Dose-response curves	N/A	N/A	N/A	Difference in plant response (growth inhibition) when exposed to self-DNA versus non-self DNA.	N/A	GO enrichment analysis, Multivariate Analysis (PCA, K-means)	N/A	no	N/A	N/A	N/A	N/A	N/A	no	Self vs. Non-Self Specificity: Only exposure to Self-DNA triggers severe root growth inhibition and stress responses (autotoxicity). Exposure to non-self DNA (from other species) does not inhibit growth, validating the autotoxic assumption in mathematical models.
Iuorio et al., 2023: Modelling how NPSF across life stages affect the spatial patterning of trees	2023	Plants - (Life stages)	Seeds S Seedlings N Adult tree A Inhibitor I (Accumulation of autotox and pathogens)	yes	yes	no	4	2 ODE (N, A) 2 PDE (S, I)	diffusion S diffusion I	NF across life stages affect patterns. JC distribution	r_P: increased mortality of N due to I r_T: sensitivity to toxicity g_A: transition rate N to A d_S: seed dispersal	LSA growth Inhibition /increased mortality: vary r_p, r_t. Seed dispersal. Growth/defense approaches	Matlab: FD FE	Neumann	no	yes	no	localized patches	stationary	yes	r_P=0. autotox-> JC distribution. Convergence to stable steady state. r_T=0 pathogens -> reduced germination. r_T=r_P=0 no JC.
Salvatori et al., 2023: Process based modelling of plants-fungus interactions explains fairy ring types and dynamics	2023	Fungus and Plants	Fungus F Plants P Fungal self-inhibitor I phytostimulants S phytotoxins T nutrients N soil water W	yes	yes	no	6	4 ODE (I, S, T, N) 2 PDE (F, P)	diffusion: F, P	Fairy rings: dynamics in space and time. Interactions F,P. classification of FR. Hydrophobicity vs phytotoxins hypothesis	a: soil water parameter s_T: sensitivity of P to T. g_F: fungal growth rate	N/A	Matlab: I.C: F=0, P=1 everywhere	Neumann	no	yes	no	ring formation	expansion	no	self-inhibition of fungus: driver of ring formation. plants–fungus interactions: drivers of the emergence of different vegetation patterns.
Bonanomi et al., 2023: Adult conspecific density effects JC patterns by modulating the recruitment exclusion zones	2023	Plants - Individual plants	N/A	no	no	IBM	N/A	N/A	N/A	connection between adult density (con- or hetero-specifics) on the probability of occurrence of JC distributions	Adult tree density NF intensity	N/A	IC: first individual in the centre Python3 MonteCarlo approach	no	no	yes	no	N/A	N/A	yes	Monospecific stands: isolated trees: JC with strong NF Forest: JC distr. Decreases as NF increases Mixed stands: rare species in heterosp: no JC co-dominated community: density of seedlings decreases with increasing of NF intensity

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Alteriis et al., 2023: Extracellular DNA secreted in yeast cultures is metabolism-specific and inhibits cell proliferation	2023	Fungi - Yeast (S. cerevisiae cultures)	Biomass Cell concentration, eDNA, Ethanol, Glucose.	N/A	N/A	Experimental / Kinetic growth model	N/A	N/A	N/A	Investigate if eDNA secretion in yeast is linked to specific metabolic states (respirative vs. fermentative) and evaluate its role in cell proliferation inhibition.	N/A	Statistical Analysis. Quantitative correlation between metabolic fluxes and eDNA accumulation.	N/A	no	N/A	N/A	N/A	N/A	N/A	no	eDNA secretion in yeast is metabolism-specific (higher during specific growth phases). Accumulated ex-self-DNA directly inhibits cell proliferation, proving that yeast cultures use self-DNA as an autotoxic signal to regulate population density.
Iuorio et al., 2024: Analysis and numerical simulations of TW due to PSNF	2024	Plants	B, T	no	yes	no	2	2 PDE (B,T)	diffusion B diffusion T. Advection T (downhill direction)	Effect of T on formation of arced bonds on Sloped terrain	b = characteristic rate of toxicity dynamics relative to plant growth rate. s: sensitivity of plants to toxicity T. m: slope parameter	LSA: bare soil saddle, Ec stable	Matlab I.C: T=0, B in centre	Neumann	no	yes	no	Localized patches Arcs	TW	no	s=0 Travelling waves in outward direction, symmetric. s>0: travelling pulses moving uphill. B peak lower. m=0 secondary rings form. m>0 Trav. Arced shape of B
Grifò et al., 2024: Far-from-equilibrium travelling pulses in sloped semi-arid environments driven by autotoxicity effects	2024	Plants	B, W, T	yes	yes	no	3	2 PDE (B,W) 1 ODE (T)	diffusion B advection W (downhill water flow extension Klausmeier)	influence of autotoxicity on migrating vegetation bands in sloped semi-arid terrains	A: precipitation parameter. H: B sensitivity to T. D: T decay rate.	LSA GSPT travelling pulse solutions	I.C: gaussian pulse for vegetation. Continuation in AUTO	Periodic	yes	no	no	localized patches	TP	no	Presence of T does not influence observed dynamics. In contrast with Iuorio, 2021 (diffference Water: advection vs diffusion)
Consolo et al., 2024: Modelling vegetation patterning on sloped terrains: The role of toxic compounds	2024	Plants	B, W, T	yes	yes	no	3	2 PDE (B,W) 1 ODE (T)	diffusion B advection W (downhill water flow extension Klausmeier)	Vegetation patterns along hillslope (downhill water flow)	A: mean annual rainfall. H: plant's sensitivity to toxicity. B: plant loss	LSA: homog. Steady state. WNA	I.C: random perturbation of homog. Steady state. Finite Element (COMSOL Multiphysics)	Periodic	yes	no	no	localized patches desert state	Oscillatory TW	no	Instability leads to oscillatory periodic patterns: TW. Toxicity-> increase of region of parameter plane where patterns and desert state arise.
Iuorio et al., 2024: Travelling waves due to NPSF in a model including tree life-stages	2024	Plants - (Life stages)	Seeds S, Seedlings N, Adult tree A, Inhibitor I (Accumulation of autotox and pathogens)	yes	yes	no	4	2 ODE (N, A) 2 PDE (S, I)	diffusion S diffusion I	JC distribution due to NF -> patterns	g_S: growth rate of S d_S: seed dispersal	LSA: bare soil unstable, E1 stable. Hopf bif. Travelling wave solutions (centre manifold reduction)	FDFE	Neumann	yes	no	no	localized patches	TW	no	travelling waves: pulled front due to NPSF
Abbas et al., 2025: PDE models for vegetation biomass and autotoxicity	2025	Plants	B, T	no	yes	no	2	2 PDE (B,T)	diffusion B diffusion T	Investigation of PSNF with different growth, inhibition, mortality (6 models)	N/A	LSA	Matlab: Implicit Euler, fix dx, dt	no	yes	no	no	Localized patches	oscillatory	no	3 classes of behaviour: 1) exp. Growth for B: unbounded increase in T. no constraints. 2) exp growth with NF that induces extra mortality: Periodic oscillations, asymmetry due to NF with a tail on a side of B distribution (depending on k). 3) stable eq. reached
Giannino, Iuorio, Soresina, 2026: Beyond water limitation in vegetation-autotoxicity patterning: a cross-diffusion model	2026	Plants - roots (healty and exposed)	Roots Re, Rh Toxicity T	no	yes	no	3	3 PDE	diffusion R (CD) diffusion T	Can NF alone induce emergence of Turing Patterns?	s: extra mortality due to T. γ: growth rate reduction due to T. σ : propagation reduction due to T	LSA: bare soil eq unstable, coexistence eq. Stable. Bifurcation analysis	Matlab: FDFE PDE2PATH	Neumann	yes	yes	yes if σ>σ_L. (only with CD)	periodic patterns spots stripes	stationary	no	Study effect of s, γ, σ. Cross-diffusion triggers Turing patterns. Double peak pulses for Biomass, with toxicity peaks located at the center of biomass gaps
Ahmad et al., 2026: Analysis and numerical simulations of fractional order model of ring formation due to PSNF	2026	Plants	B, T	yes	yes	no	2	1 ODE (T) 1 PDE (B) time-fract. (Caputo)	Diffusion B	Evaluate memory effects (fractional derivative) on fairy ring formation and dynamics.	k: derivation index. a: plant basal mortality rate relative to growth rate	LSA: unique eq. stability depends on a.	Matlab: MOL, FD, PECE solver	Neumann	yes	yes	no	ring formation	invasion fronts TW	no	Results match Cartenì, 2012. As k decreases the area of spread of B, totB increase, totT, maxB, maxT decrease. The width of rings increase, the internal ring diameter decrease, slower dynamics.
Consolo et al., 2026: How inertia affects autotoxicity-mediated vegetation dynamics: from close-to to far-from-equilibrium patterns	2026	Plants	B, W, T, J (veg. Flux density)	yes	yes	no	4	1 ODE (T) 3 PDE (B, W, J) (hyperb.)	advection U (slope terrain)	Role of Inertia in pattern dynamics	A: rainfall parameter H: plant's sensitivity to toxicity. t: vegetation inertial time	LSA WNA GSPT	COMSOL Multiphysics Continuation in AUTO	Periodic	yes	no	no	Localized patches	Oscillatory	no	(A, B) parameter plane divided in 3 region. Oscillating patterns, migration speed increase with A (uphill direction). Inertia affect the region of patterns